AN ALMOST EVERYWHERE DIVERGENT FOURIER SERIES OF THE CLASS $L(\log^+ \log^+ L)^{1-\varepsilon}$

Y. M. CHEN

It is well known that there exists an almost everywhere divergent Fourier series of the class L, but none is known of the class $L \log^+ L$.† In this paper we consider the gap between L and $L \log^+ L$. We shall construct a function $f(x)$ of the class $L(\log^+ \log^+ L)^{1-\varepsilon}$ for any given $\varepsilon > 0$, such that the Fourier series of f diverges almost everywhere.

To suit the double $\log^+$, we use $\log^+ f, f \geqslant 0$, to denote $\log f$ if $f > e$ and $\log^+ f = 1$ otherwise. This is slightly different from the usual notation that $\log^+ f = \log f$ when $f > 1$ and $\log^+ f = 0$ otherwise.

LEMMA 1. *There exists a sequence of functions $\phi_n(x)$ satisfying the following conditions*:

(i) $\phi_n(x) \geqslant 0, \int_0^{2\pi} \phi_n(x)\,dx = 2 \quad (n = 1, 2, \ldots)$;

(ii) *$\phi_n(x)$ is a function of bounded variation*;

(iii) *there exists an absolute constant A such that*

(a) $|S_p(x, \phi_n)| < A \log n \quad (p = 1, 2, \ldots)$ *for all $x \in H_n$, where $S_p(x, \phi_n)$ is the p-th partial sum of the Fourier series of ϕ_n and $H_n \subset [0, 2\pi]$ will be defined later. Furthermore, H_n satisfies*

(b) $\lim_{n \to \infty} mH_n = 2\pi$;

(iv) *for any $\varepsilon' > 0$, there exist a positive constant $\alpha = \alpha(\varepsilon')$, an integer $N = N(\varepsilon')$, and a set $\mathscr{E}_n \subset [0, 2\pi]$ such that*

(a) $m\mathscr{E}_n > 2\pi - \varepsilon'$, *whenever $n > N$*;

(b) *for any $x \in \mathscr{E}_n$, there is a positive integer p_x depending on x, such that*

$$|S_{p_x}(x, \phi_n)| > \alpha \log n \quad (n > N);$$

(c) *the integer p_x satisfies $n \leqslant p_x \leqslant m_n$, where m_n depends on n but not on ε'*;

(d) $m_n = m_{k,n}$ *when $k = n$, where*

$$2m_{k,n} + 1 = 3^k(2n+1)^3 \quad (k = 1, 2, \ldots, n).$$

The numbers $m_{k,n}$ will be used in defining the functions ϕ_n.

The existence of the functions ϕ_n satisfying all the conditions except (iv) has been established by A. N. Kolmogorov. In order to establish our Lemma 1, we shall follow the argument in N. K. Bary [**1**; pp. 391–402] (pp. 430–437 in Vol. 1 of the English translation). Kolmogorov's functions have to be modified to a certain extent in order to satisfy the new conditions.

Received 6 September, 1967; revised 14 February, 1968

† For an up-to-date detailed discussion of this problem, we refer to P. L. Ul'ynov [**10, 11**] and N. K. Bary [**1**]; and also [**3, 7, 6, 5, 2**] for recent advance concerning the class L^p $(p > 1)$.

Let

$$A_{k,n} = \frac{4k\pi}{2n+1}, \quad \Delta_{k,n} = \left(A_{k,n} - \frac{1}{m_{k,n}}, \; A_{k,n} + \frac{1}{m_{k,n}}\right) \quad (k = 1, 2, \ldots, n),$$

$$\phi_n(x) = \begin{cases} m_{k,n}/n & x \in \Delta_{k,n} \\ 0 & \text{outside of all } \Delta_{k,n} \end{cases} \qquad (k = 1, 2, \ldots, n).$$

Here it is obvious that each ϕ_n is a function of bounded variation. This is condition (ii) of Lemma 1. It also follows from the definition of $\phi_n(x)$ that the condition (i) of Lemma 1 is satisfied:

$$\int_0^{2\pi} \phi_n(x)\,dx = \sum_{k=1}^{n} \int_{\Delta_{k,n}} \phi_n(x)\,dx = n \cdot \frac{2}{n} = 2.$$

For $n \geqslant 2$, let

$$d_{k,n} = \left[A_{k,n} + \frac{1}{n\sqrt{\log n}}, \; A_{k+1,n} - \frac{1}{n\sqrt{\log n}}\right] \quad (k = 1, 2, \ldots, n-1),$$

and

$$H_n = \bigcup_{k=1}^{n-1} d_{k,n}.$$

Then we have

$$mH_n = (n-1)\left[\frac{4\pi}{2n+1} - \frac{2}{n\sqrt{\log n}}\right] \to 2\pi,$$

as $n \to \infty$. The condition (iii) (b) of Lemma 1 is satisfied.

Now with reference of Fig. 1 and Fig. 2, it is easy to see that for any integers k and j $(1 \leqslant k \leqslant n-1, 1 \leqslant j \leqslant n)$, if $x \in d_{k,n}$ and $t \in \Delta_{j,n}$, $n \geqslant 2$,

$$|t-x| > \frac{1}{n\sqrt{\log n}} - \frac{1}{4n^3} > \frac{1}{2n\sqrt{\log n}}. \tag{1.1}$$

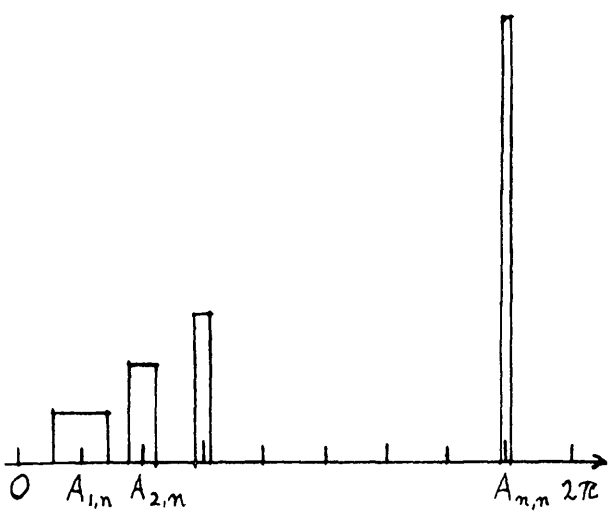

Fig. 1

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

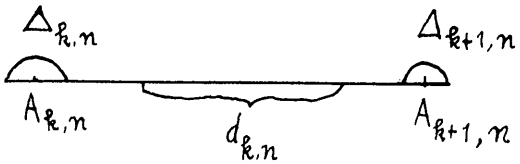

Fig. 2

Moreover, for $j < k$, $t \in \Delta_{j,n}$ and $x \in d_{k,n}$,

$$\begin{aligned}|t-x| &= x-t = (x-A_{k,n})+(A_{k,n}-A_{j,n})+(A_{j,n}-t)\\ &\geqslant \left(\frac{1}{n\sqrt{\log n}}-\frac{1}{m_{k,n}}\right)+\frac{4\pi(k-j)}{2n+1}-\frac{1}{m_{j,n}}\\ &> \frac{4\pi(k-j)}{2n+1}+\frac{1}{n\sqrt{\log n}}-\frac{2}{m_{j,n}}\\ &> (k-j)\frac{4\pi}{2n+1} \quad (n \geqslant 3);\end{aligned} \tag{1.2}$$

and for $j > k+1$,

$$\begin{aligned}|t-x| &= t-x = (t-A_{j,n})+(A_{j,n}-A_{k+1,n})+(A_{k+1,n}-x)\\ &> (j-k-1)\frac{4\pi}{2n+1}+\frac{1}{n\sqrt{\log n}}-\frac{3}{m_{k+1,n}}\\ &> \frac{4\pi(j-k-1)}{2n+1} \quad (n \geqslant 3).\end{aligned} \tag{1.3}$$

We come now to the proof of the condition (iii) (a):

$$|S_p(x, \phi_n)| > A \log n \quad (p = 1, 2, \ldots),$$

for all $x \in H_n$. The inequality (1.1) will be necessary for the proof. We have to estimate the expression

$$S_p(x, \phi_n) = \frac{1}{\pi}\int_0^{2\pi} \phi_n(t)\,\frac{\sin(p+\frac{1}{2})(t-x)}{2\sin[(t-x)/2]}\,dt,$$

for $x \in d_{k,n}$, where $S_p(x, \phi_n)$ is the p-th partial sum of the Fourier series of $\phi_n(x)$.

To this end, it is enough to consider the modified partial sum:

$$\tilde{S}_p(x, \phi_n) = \frac{1}{\pi}\int_0^{2\pi} \phi_n(t)\,\frac{\sin p(t-x)}{t-x}\,dt, \tag{1.4}$$

since $\int_0^{2\pi} \phi_n(t)\,dt = 2$, and so $|S_p(x, \phi_n)-\tilde{S}_p(x, \phi_n)| \leqslant K$, where K is independent of n. We have

$$|\tilde{S}_p(x, \phi_n)| \leqslant \frac{1}{\pi}\int_0^{2\pi} \frac{\phi_n(t)}{|t-x|}\,dt = \frac{1}{\pi}\sum_{j=1}^{n}\int_{\Delta_{j,n}} \frac{\phi_n(t)}{|t-x|}\,dt. \tag{1.5}$$

From the relations (1.1) and $\int_{\Delta_{j,n}} \phi_n(t)\,dt = 2/n$ $(j = 2, \ldots, n)$, we see that if $x \in d_{k,n}$, $j = k$ or $j = k+1$, then

$$\int_{\Delta_{j,n}} \frac{\phi_n(t)}{|t-x|}\,dt \leqslant \frac{2}{n}\{(2n\sqrt{\log n})^{-1}\}^{-1} = 4\sqrt{\log n}.$$

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

On the other hand, when $x \in d_{k,n}$, by (1.2) and (1.3) we have

$$\int_{\Delta_{j,n}} \frac{\phi_n(t)}{|t-x|} dt \leqslant \begin{cases} \dfrac{2n+1}{4\pi} \cdot \dfrac{1}{k-j} \cdot \dfrac{2}{n} < \dfrac{2}{k-j} & (j < k), \\ \dfrac{2n+1}{4\pi} \cdot \dfrac{1}{j-k-1} \cdot \dfrac{2}{n} < \dfrac{2}{j-k-1} & (j > k+1). \end{cases}$$

It follows that

$$|\tilde{S}_p(x, \phi_n)| \leqslant \sum_{j=1}^{k-1} \frac{2}{k-j} + 4\sqrt{\log n} + \sum_{j=k+2}^{n} \frac{2}{j-k-1}$$

$$= \left[4\sqrt{(\log n)} + 2 \sum_{j=1}^{k-1} \frac{1}{j} + 2 \sum_{j=1}^{n-k-1} \frac{1}{j} \right] < 8 \log n; \tag{1.6}$$

whence

$$|S_p(x, \phi_n)| < A \log n, \tag{1.7}$$

where A is an absolute constant and $x \in \bigcup_{k=1}^{n-1} d_{k,n} = H_n$. So the conditions (i), (ii), (iii) of Lemma 1 are satisfied.

It remains to prove condition (iv) of Lemma 1, for which the proof is longer. We shall now prove that

$$\frac{1}{\pi} \int_{\bigcup_{j=1}^{k-1} \Delta_{j,n}} \phi_n(t) D_{m_{k,n}}(t-x) \, dt = o(\log n), \tag{1.8}$$

as $n \to \infty$, where $D_{m_{k,n}}(t)$ is the Dirichlet kernel of order $m_{k,n}$ and $x \in d_{k-1,n}$, $2 \leqslant k \leqslant n$.

In fact, by virtue of the Riemann–Lebesgue theorem the left member in (1.8) does not exceed $V/m_{k,n}$, where $V = V\left(\bigcup_{j=1}^{k-1} \Delta_{j,n}\right)$ is the total variation of the function $\dfrac{K\phi_n(t)}{\sin[(t-x)/2]}$ over $\bigcup_{j=1}^{k-1} \Delta_{j,n}$. Here $t-x < 0$ for $t \in \Delta_{j,n}$ $(j = 1, 2, \ldots, k-1)$ and $x \in d_{k-1,n}$. The function $|\sin(t-x)/2|^{-1}$ is monotonically increasing while $\phi_n(t)$ is constant in each $\Delta_{j,n}$ and

$$|\sin[(t-x)/2]|^{-1} \leqslant \frac{\pi}{|t-x|} \leqslant \frac{K_n}{k-j} \quad (1 \leqslant j \leqslant k-2).$$

But when $t \in \Delta_{k-1,n}$ and $x \in d_{k-1,n}$,

$$|t-x| \geqslant \frac{1}{n\sqrt{\log n}} - \frac{1}{m_{k-1,n}} > \frac{1}{2n\sqrt{\log n}},$$

and

$$|\sin[(t-x)/2]|^{-1} \leqslant Kn(\log n)^{\frac{1}{2}},$$

where $|\sin[(t-x)/2]|^{-1}$ is decreasing in $\Delta_{k-1,n}$ for fixed x. It follows that

$$V \leqslant K_1 \sum_{j=1}^{k-2} \frac{m_{j,n}}{n} \cdot \frac{n}{k-j} + K_2 \frac{m_{k-1,n}}{n} \cdot n (\log n)^{\frac{1}{2}},$$

where the two constants K_1 and K_2 on the right may be chosen so that they are

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

independent of $x \in d_{k-1,n}$ $(2 \leqslant k \leqslant n)$. Hence

$$V \leqslant o(m_{k,n} \log n) + K \sum_{j=1}^{k-2} \frac{m_{j,n}}{k-j}$$

$$= o(m_{k,n} \log n) + O\left(\sum_{j=1}^{k-2} m^3 3^j\right)$$

$$= o(m_{k,n} \log n),$$

as $n \to \infty$. This implies (1.8).

Next, for the Dirichlet kernel $D_{m_{k,n}}$ we have the relation:

$$|D_{m_{k,n}}(t-x) - D_{m_{k,n}}(A_{j,n}-x)| \leqslant K\left\{\frac{n}{(j-k)^2}\right\} + K, \tag{1.9}$$

when $2 \leqslant k \leqslant n$, $x \in d_{k-1,n}$, $t \in \Delta_{j,n}$, $j = k+1, \ldots, n$. In fact, as $n \to \infty$,

$$D_{m_{k,n}}(t-x) - D_{m_{k,n}}(A_{j,n}-x) = \frac{\sin m_{k,n}(t-x)}{t-x} - \frac{\sin m_{k,n}(A_{j,n}-x)}{A_{j,n}-x} + O(1)$$

$$= \frac{2}{t-x}\left\{\cos\frac{m_{k,n}}{2}(t-A_{j,n}-2x)\sin\frac{m_{k,n}}{2}(t-A_{j,n})\right\}$$

$$+ \frac{A_{j,n}-t}{(t-x)(A_{j,n}-x)} \sin m_{k,n}(A_{j,n}-x) + O(1)$$

$$= R_1 + R_2 + O(1), \text{ say.} \tag{1.10}$$

When $x \in d_{k-1,n}$, $t \in \Delta_{j,n}$ $(j > k)$, $|t-x| > K(j-k)/n$, $|t-A_{j,n}| \leqslant m_{j,n}^{-1}$, and so

$$|R_1| \leqslant K\left\{\frac{n}{j-k} \cdot \frac{m_{k,n}}{m_{j,n}}\right\} \leqslant K\left\{\frac{n.3^{-(j-k)}}{(j-k)}\right\} \leqslant K\left\{\frac{n}{(j-k)^2}\right\}.$$

Since $|A_{j,n}-x| > K(j-k)/n$ when $j \geqslant k+1$, $x \in d_{k-1,n}$, $t \in \Delta_{j,n}$; we have

$$|R_2| \leqslant K.\frac{n}{j-k}.m_{j,n}^{-1}.\frac{n}{j-k} \leqslant K(j-k)^{-2}$$

Substituting the inequalities relating to R_1 and R_2 in (1.10), we obtain (1.9).

We shall now prove that, when $x \in d_{k-1,n}$ and $2 \leqslant k \leqslant n$,

$$\frac{1}{\pi} \int_{\bigcup_{j=k}^{n} \Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(t-x)\, dt = \frac{1}{\pi} \int_{\bigcup_{j=k}^{n} \Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(A_{j,n}-x)\, dt + o(\log n), \tag{1.11}$$

as $n \to \infty$. We consider separately cases $j \geqslant k+1$ and $j = k$.

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

For the first case, we use (1.9):

$$\frac{1}{\pi}\int_{\bigcup_{j=k+1}^{n}\Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(t-x)\,dt$$

$$=\frac{1}{\pi}\int_{\bigcup_{j=k+1}^{n}\Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(A_{j,n}-x)\,dt$$

$$+O\left\{\int_{\bigcup_{j=k+1}^{n}\Delta_{j,n}} \phi_n(t)\,dt+\sum_{j=k+1}^{n}\frac{n}{(j-k)^2}\int_{\Delta_{j,n}}\phi_n(t)\,dt\right\}$$

$$=\frac{1}{\pi}\int_{\bigcup_{j=k+1}^{n}\Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(A_{j,n}-x)\,dt+O(1)\quad (x\in d_{k-1,n}).$$

For the second case, $j=k$, $x\in d_{k-1,n}$,

$$|t-x|>\frac{1}{n(\log n)^{\frac{1}{2}}}-\frac{1}{m_{k,n}}>\frac{1}{2n(\log n)^{\frac{1}{2}}}\quad (t\in\Delta_{j,n},\ x\in d_{k-1,n}),$$

and

$$|A_{j,n}-t|<\frac{1}{m_{k,n}},\quad |A_{j,n}-x|>\frac{1}{n(\log n)^{\frac{1}{2}}}.$$

This implies that R_1 and R_2 in (1.10) satisfy respectively

$$|R_1|\leqslant n(\log n)^{\frac{1}{2}},\quad |R_2|\leqslant n^2\log n.m_{k,n}^{-1}=o(1),$$

as $n\to\infty$. It follows that

$$\frac{1}{\pi}\int_{\Delta_{k,n}}\phi_n(t)\,D_{m_{k,n}}(t-x)\,dt=\frac{1}{\pi}\int_{\Delta_{k,n}}\phi_n(t)\,D_{m_{k,n}}(A_{k,n}-x)\,dt+O\{(\log n)^{\frac{1}{2}}\},$$

as $n\to\infty$. Combining the separate cases, we obtain (1.11).

Next, it follows from (1.8) and (1.11) that if $2\leqslant k\leqslant n$ and $x\in d_{k-1,n}$, then

$$\frac{1}{\pi}\int_{\bigcup_{j=1}^{n}\Delta_{j,n}}\phi_n(t)\,D_{m_{k,n}}(t-x)\,dt=\frac{1}{\pi}\int_{\bigcup_{j=k}^{n}\Delta_{j,n}}\phi_n(t)\,D_{m_{k,n}}(A_{j,n}-x)\,dt+o(\log n),\quad (1.12)$$

as $n\to\infty$. Here

$$D_{m_{k,n}}(A_{j,n}-x)=\frac{\sin(m_{k,n}+\frac{1}{2})(A_{j,n}-x)}{2\sin[(A_{j,n}-x)/2]}=\frac{\sin(2m_{k,n}+1)\left(\frac{2j\pi}{2n+1}-\frac{x}{2}\right)}{2\sin[(A_{j,n}-x)/2]}$$

$$=-\frac{\sin(2m_{k,n}+1)\frac{x}{2}}{2\sin[(A_{j,n}-x)/2]},\quad (1.13)$$

since $(2m_{k,n}+1)=3^k(2n+1)^3=(2n+1)\lambda_{k,n}$, where $\lambda_{k,n}$ is an integer.

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Again, for $2 \leqslant k \leqslant j \leqslant n$ and $x \in d_{k-1,n}$,

$$0 \leqslant A_{j,n} - x \leqslant A_{j,n} - A_{k-1,n} = (j-k+1)\frac{4\pi}{2n+1} < 2\pi.$$

So we have for $x \in d_{k-1,n}$, $2 \leqslant k \leqslant j \leqslant n$,

$$|D_{m_{k,n}}(A_{j,n}-x)| \geqslant \left|\sin(2m_{k,n}+1)\frac{x}{2}\right| \cdot \frac{2n+1}{4\pi}\left(\frac{1}{j-k+1}\right). \tag{1.14}$$

Combining (1.12), (1.13) and (1.14), together with $\int_{\Delta_{j,n}} \phi_n(t)\,dt = 2/n$, we obtain the following inequality:

$$\frac{1}{\pi}\left|\int_{\bigcup_{j=1}^{n} \Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(t-x)\,dt\right|$$

$$\geqslant \frac{1}{\pi} \cdot \frac{2n+1}{4\pi}\left(\frac{2}{n}\right)\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| \sum_{j=k}^{n} \frac{1}{j-k+1} + o(\log n)$$

$$\geqslant \frac{1}{\pi^2}\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| \sum_{j=k}^{n} \frac{1}{j-k+1} + o(\log n), \tag{1.15}$$

as $n \to \infty$, when $x \in d_{k-1,n}$ and $2 \leqslant k \leqslant n$.

Moreover, if we restrict k so that $k < n - \sqrt{n}$, then $n-k > \sqrt{n}$, and

$$\sum_{j=k}^{n} \frac{1}{j-k+1} = \sum_{1}^{n-k+1} \frac{1}{j} > \log\sqrt{n} = \tfrac{1}{2}\log n.$$

Consequently, with this new restriction for k, we have

$$\frac{1}{\pi}\left|\int_{\bigcup_{j=1}^{n} \Delta_{j,n}} \phi_n(t)\, D_{m_{k,n}}(t-x)\,dt\right| \geqslant \frac{1}{2\pi^2}\log n\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| + o(\log n),$$

as $n \to \infty$, for $x \in d_{k-1,n}$. This means that for $x \in d_{k-1,n}$ and $2 \leqslant k < n - \sqrt{n}$,

$$|S_{m_{k,n}}(x, \phi_n)| > \frac{1}{2\pi^2}\log n\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| + o(\log n), \tag{1.16}$$

as $n \to \infty$.

For given $\delta > 0$, let $G_{k-1,n}(\delta)$ denote the set of all $x \in d_{k-1,n}$ for which

$$\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| < \delta.$$

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Let $G_n(\delta) = \bigcup_{k=1}^{n-1} G_{k,n}(\delta)$. We shall now show that $mG_n(\delta) \leqslant K\delta$, where K is independent of n. To see this, if $x \in d_{k-1,n}$, $2 \leqslant k \leqslant n$, then

$$\frac{A_{k-1,n}}{2} < \frac{x}{2} < \frac{A_{k,n}}{2}$$

i.e.

$$(k-1)\frac{2\pi}{2n+1} < \frac{x}{2} < k.\frac{2\pi}{2n+1}.$$

Write $2m_{k,n}+1 = 3^k(2n+1)^2(2n+1) = \lambda_{k,n}(2n+1)$. Then

$$\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| = \left|\sin\lambda_{k,n}(2n+1)\left[\frac{x}{2}-(k-1).\frac{2\pi}{2n+1}\right]\right|$$

and it suffices to consider the behaviour of

$$|\sin\lambda_{k,n}(2n+1)\,t|,$$

for $0 < t < 2\pi/(2n+1)$, or $|\sin\lambda_{k,n}u|$ for $(2n+1)\,t = u$, $0 < u < 2\pi$. But the set of all u for which $|\sin\lambda_{k,n}u| < \delta$ consists of $4\lambda_{k,n}$ intervals, the length of each of which is $\arcsin\delta/\lambda_{k,n}$, and therefore the total measure of such intervals does not exceed $4\arcsin\delta$ which is independent of n. This implies that the measure of the set of x,

$$(k-1)\frac{2\pi}{2n+1} < \frac{x}{2} < \frac{2k\pi}{2n+1},$$

such that

$$\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| < \delta,$$

is less than

$$\frac{8\arcsin\delta}{2n+1} < \frac{4\arcsin\delta}{n}.$$

It follows that $\qquad mG_{k-1,n}(\delta) < 4\arcsin\delta/n,$

and consequently, $\qquad mG_n(\delta) < 4\arcsin\delta \leqslant K\delta.$

This means that ir $x \notin G_n(\delta)$, $x \in d_{k-1,n}$, $2 \leqslant k \leqslant n$, then

$$\left|\sin(2m_{k,n}+1)\frac{x}{2}\right| \geqslant \delta. \tag{1.17}$$

From (1.16) and (1.17), if $x \in d_{k-1,n}$, $x \notin G_n(\delta)$ and if $2 \leqslant k \leqslant n-\sqrt{n}$, then

$$|S_{m_{k,n}}(x, \phi_n)| \geqslant \frac{\delta}{2\pi^2}\log n - o(\log n).$$

Let

$$E_n = \bigcup_{k=1}^{[n-\sqrt{n}]-1} d_{k,n},$$

and let $\mathscr{E}_n = [\mathscr{C}G_n(\delta)] \cap E_n$, where $\mathscr{C}G_n(\delta) = \{[0, 2\pi]\setminus G_n(\delta)\}$ is the complement

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

of $G_n(\delta)$. Then

$$mE_n = \{[n-\sqrt{n}]-1\}\left(\frac{4\pi}{2n+1} - \frac{2}{n\sqrt{\log n}}\right) \to 2\pi,$$

as $n \to \infty$.

It follows that $m\mathscr{E}_n \geqslant 2\pi - o(1) - O(\delta) > 2\pi - \varepsilon'$, if δ is suitably chosen and $n > N$, where N is sufficiently large. This is the condition (iv) (a).

Since $\mathscr{E}_n \subset E_n$, if $x \in \mathscr{E}_n$ then $x \in E_n$ and so a $k = k(n)$ can be found such that $x \in d_{k-1,n}$, $k \leqslant n-\sqrt{n}$, $x \notin G_n(\delta)$, together with

$$|S_{m_k,n}(x, \phi_n)| \geqslant \frac{\delta}{2\pi^2} \log n - o(\log n), \text{ as } n \to \infty,$$

and $m_{1,n} \leqslant m_{k,n} \leqslant m_n = m_{n,n}$ (*cf.* condition (iv) (d)).

In view of $m_{1,n} \geqslant n$, we can assert that, for any $x \in \mathscr{E}_n$ there is a p_x, $n \leqslant p_x \leqslant m_n$ such that

$$|S_{p_x}(x, \phi_n)| \geqslant \frac{\delta}{2\pi^2} \log n - o(\log n),$$

as $n \to \infty$, and $m\mathscr{E}_n > 2\pi - \varepsilon'$ for $n > N$. If we take n sufficiently large, then

$$|S_{p_x}(x, \phi_n)| > \frac{\delta}{4\pi^2} \log n = \alpha \log n,$$

for $n > N$. This proves the conditions (iv) (b) and (iv) (c). The proof of Lemma 1 is now complete.

2. We come now to the construction of our function $f(x)$. For this, we shall follow the argument in Bary [1] (English translation, Vol. 1, pp. 438–442).*

We suppose that $0 < \varepsilon \leqslant \frac{1}{2}$, and we choose the sequence $\{n_k\}$ such that $\sum(\log n_k)^{-\varepsilon} < \infty$ (which implies that $\sum(\log n_k)^{-\frac{1}{2}} < \infty$). Let

$$f(x) = \sum_{k=1}^{\infty} \frac{\phi_{n_k}(x)}{\log n_k}. \tag{2.1}$$

Then $f(x) \in L(\log^+ \log^+ L)^{1-\varepsilon}$, for the given $\varepsilon > 0$, and the Fourier series of $f(x)$ diverges almost everywhere.

The proof of the almost everywhere divergence of the Fourier series of $f(x)$ is the same as in [1], for this proof requires only parts (i)–(iii) and (iv) (a)–(c) of Lemma 1, and these parts form Bary's Lemma on p. 430 of [1] (English translation).

It remains to show that $f(x) \in L(\log^+ \log^+ L)^{1-\varepsilon}$, for which we require the following Lemma 2.

LEMMA 2. *Let* $g(x) = \sum_1^{\infty} g_n(x)$, *where* $g_n(x) \geqslant 0$ $(n = 1, 2, \ldots)$ *are measurable functions. If* $g_n(x) > n$ *whenever* $g_n(x) \neq 0$, *and if*

$$\sum_{n=1}^{\infty} \int_0^{2\pi} g_n(x)[\log^+ \log^+ g_n(x)]^{1-\varepsilon} dx < \infty, \tag{2.2}$$

for an arbitrary $\varepsilon > 0$, *then* $g \in L(\log^+ \log^+ L)^{1-\varepsilon}$.

* Here it should be noted that our function $f(x)$ is a new function. Our result cannot be included in the results in [1].

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Proof of Lemma 2. Write

$$\psi(x) = g(x)\{[\log^+ \log^+ g(x)]^{1-\varepsilon}/\log^+ g(x)\}. \tag{2.3}$$

Since

$$\frac{[\log^+ \log^+ g]^{1-\varepsilon}}{\log^+ g} \leqslant \frac{[\log^+ \log^+ g_n]^{1-\varepsilon}}{\log^+ g_n} \quad (n = 1, 2, \ldots),$$

$$\psi(x) \leqslant \sum_1^\infty g_n(x)\{[\log^+ \log^+ g_n]^{1-\varepsilon}/\log^+ g_n\} = \sum_1^\infty \Phi_n = \Phi(x),$$

say. We shall make use of Zygmund's inequality.† From the relation

$$\begin{aligned} n^2 \int_0^{2\pi} |\overline{\Phi}_n(x)|\, dx &\leqslant A.n^2 \int_0^{2\pi} \Phi_n \log^\oplus n^2 \Phi_n\, dx + B \\ &\leqslant n^2 A \int_0^{2\pi} \Phi_n \log^+ \Phi_n\, dx + 2n^2 A \int_0^{2\pi} (\log n)\, \Phi_n\, dx + B \\ &\leqslant 3n^2 A \int_0^{2\pi} \Phi_n(x) \log^+ \Phi_n(x)\, dx + B \\ &= n^2 A' \int_0^{2\pi} \Phi_n(x) \log^+ \Phi_n(x)\, dx + B, \end{aligned}$$

we find

$$\int_0^{2\pi} |\overline{\Phi}_n(x)|\, dx \leqslant A' \int_0^{2\pi} \Phi_n(x) \log^+ \Phi_n(x)\, dx + B/n^2,$$

where A, A', B are positive absolute constants. It follows that

$$\begin{aligned} \int_0^{2\pi} |\overline{\Phi}(x)|\, dx &\leqslant \sum_1^\infty \int_0^{2\pi} |\overline{\Phi}_n(x)|\, dx \\ &\leqslant A' \sum_1^\infty \int_0^{2\pi} \Phi_n(x) \log^+ \Phi_n(x)\, dx + B' \\ &\leqslant A'' \sum_1^\infty \int_0^{2\pi} g_n(x)[\log^+ \log^+ g_n(x)]^{1-\varepsilon}\, dx + B' < \infty, \end{aligned}$$

where A'' and B' are also absolute constants. Since both Φ and $\overline{\Phi}$ are L-integrable and $\Phi \geqslant 0$, it follows from the Riesz theorem‡ that $\Phi \in L \log^+ L$. This implies that $g(x) \in L(\log^+ \log^+ L)^{1-\varepsilon}$ and the proof of Lemma 2 is complete.

---

† A. Zygmund: Sur les fonctions conjuguées. *Fund. Math.*, 13 (1929), 284–303; 18 (1932) p.312; *cf.* also [**1**; Vol. 2 (English translation), p. 108]. Here we use $\log^\oplus f$ to denote, as the usual notation $\log^+ f$, that $\log^\oplus f = \log f$ when $f > 1$, and zero otherwise, in order distinguish between it and our symbol $\log^+ f$ used here.

‡ M. Riesz: Sur les fonctions conjuguées, *Math. Z.*, 27 (1927) 218–244; see also [**1**; Vol. 2 (English translation), p. 112].

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Finally, to show that $f(x) \in L(\log^+ \log^+ L)^{1-\varepsilon}$, it is sufficient to apply Lemma 2. Write

$$f(x) = \sum_{k=1}^{\infty} \phi_{n_k}(x)/\log n_k = \sum_{1}^{\infty} f_k(x),$$

say, where

$$f_k(x) \geqslant \frac{n_k^3}{\log n_k} > k,$$

when $f_k(x) \neq 0$. We have

$$\begin{aligned}
&\sum_{k=1}^{\infty} \int_0^{2\pi} f_k(x)[\log^+ \log^+ f_k(x)]^{1-\varepsilon}\, dx \\
&\leqslant \sum_{k=1}^{\infty} \int_0^{2\pi} \frac{\phi_{n_k}(x)}{\log n_k} [\log^+ \log^+ \phi_{n_k}(x)]^{1-\varepsilon}\, dx \\
&\leqslant \sum_{k=1}^{\infty} \int_0^{2\pi} \frac{\phi_{n_k}(x)}{\log n_k} \left(\log^+ \log^+ \frac{3^{n_k}(2n_k+1)^3}{n_k}\right)^{1-\varepsilon} dx \\
&< 2\sum_{k=1}^{\infty} (\log n_k)^{-\varepsilon} \int_0^{2\pi} \phi_{n_k}(x)\, dx = 4\sum_{k=1}^{\infty} (\log n_k)^{-\varepsilon} < \infty. \qquad (2.4)
\end{aligned}$$

Hence $f(x) \in L(\log^+ \log^+ L)^{1-\varepsilon}$. This completes the construction of our function $f(x)$.

Remark 1. The Fourier series of the function $f(x)$ we have constructed is somewhat similar to Marcinkiewicz's example [**8**], i.e.

$$\limsup_{n\to\infty} S_n(x, f) < +\infty, \quad \text{almost everywhere.}$$

It is easy to see that, by slight modification of the example, we can change it to a Fourier series similar to Kolmogorov's divergent Fourier series, i.e.

$$\limsup_{n\to\infty} S_n(x, f) = +\infty, \quad \text{almost everywhere.}$$

Remark 2. In a previous paper [**4**], the author has conjectured that there exists a function $f \in L(\log^+ \log^+ L)$, such that the conjugate function $\bar{f}(x) \in L$ and the Fourier series of f diverges everywhere. As a matter of fact, the result given here is very near to the ideal function.

On the other hand, A. Zygmund† has conjectured that there exists a function $f \in L$, such that its Fourier series diverges almost everywhere at the speed of $o(\log n)$. As a "counter balance" of this conjecture, it is equivalent to that of the existence of a function $f(x) \in L \log^+ L$, such that its Fourier series diverges almost everywhere.

It has been thought that Zygmund's conjecture has been solved by E. M. Stein.‡ E. M. Stein [**9**; Theorem 6] has proved that there exists an integrable function $f(x)$

† For a discussion of this problem see [**4**] and [**11**].

‡ *Cf.* P. L. Ul'ynov [**11**; p. 32, footnote (p. 27, footnote of the English translation)].

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

such that

$$\sup_{\substack{m,n \\ m-n=k}} S_n(x,f) - S_m(x,f) = O\{\lambda(k)\log k\}, \quad \text{as } k \to \infty,$$

is false for almost every x, where $\lambda(m)$ is any sequence decreasing to zero as $m \to \infty$.

However, Stein's result does not imply that it is still true if we replace n by a fixed constant, say $n = 0$, and Zygmund's conjecture does not follow from Stein's theorem. The problem remains open.

I wish to express my hearty thanks to the referee for some improvements in the presentation of this paper.

References

1. N. K. Bary, *Trigonometric series* (Moscow, 1961). Also English translation by M. F. Mullins, " A treatise on trigonometric series ", Vol. 1 and Vol. 2 (1964).
2. P. Billard, " Sur la convergence presque partout des séries de Fourier-Walsh des fonctions de l'espace $L^2(0, 1)$ ", *Studia Math.*, 28 (1966–1967), 363–388.
3. L. Carleson, " On convergence and growth of partial sums of Fourier series ", *Acta Math.*, 116 (1966), pp. 135–157.
4. Y. M. Chen, " A remarkable divergent Fourier series ", *Proc. Japan Acad.*, 38, 239–244 (1962).
5. R. A. Hunt, " On the convergence of Fourier series ", *Notices Amer. Math. Soc.*, June 1967, Vol. 14, No. 4, p. 537.
6. J. P. Kahane and Y. Katznelson, " Sur les ensembles de divergence des séries trigonométriques ", *Studia Math.*, 36 (1966), 305–306.
7. A. Maté, *Convergence of L^2 Fourier series* (detailed exposition of Carleson's proof). Lecture Note, Bolyai Institute Univ. of Szeged (Hungary, 1967).
8. J. Marcinkiewicz, " Sur les séries de Fourier ", *Fund. Math.*, 27 (1936), 38–69.
9. E. M. Stein, " On limits of sequences of operators ", *Annales Math.*, 74 (1961), 140–170.
10. P. L. Ul'ynov, " On divergence of Fourier series " (Russian), *Uspehi Mat. Nauk*, 12 (1957), 75–132.
11. P. L. Ul'ynov, " Solved and unsolved problems in the theory of trigonometric and orthogonal series ", *Uspehi Mat. Nauk.*, 19 (1964), 3–69; also English translation by S. Verblunsky, *Russian Math. Surveys*, 19 (1964), 1–62.

Department of Mathematics,
University of Hong Kong.

14697750, 1969, 1, Downloaded from https://londmathsoc.onlinelibrary.wiley.com/doi/10.1112/jlms/s1-44.1.643 by Peking University Health, Wiley Online Library on [01/03/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License